

Sri Datta Ganesh Bandreddi

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EDUCATION

CARNEGIE MELLON UNIVERSITY

Master of Science in AI Engineering, Biomedical Engineering | GPA: 4.00/4.00

Pittsburgh, PA

Dec 2026

VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY

Bachelor of Technology, Mechanical Engineering | GPA: 8.44/10.00

Nagpur, India

Jul 2022

SKILLS

Languages: Python, C++, Go, Java, SQL, Bash /Shell Scripting

GPU (Acceleration): CUDA, cuDNN, cuBLAS, TensorRT, Triton Inference Server, NCCL, Nsight Systems/Compute

DL Frameworks: PyTorch, JAX, TensorFlow, HF, NeMo/Megatron-LM, LoRA/QLoRA, FP8/BF16 Mixed Precision

Data & ETL: Rapids (cuDF, cuML), NVTabular, DALI, NumPy, SciPy, Pandas, SparkML

Systems & MLOps: Docker, Kubernetes, Slurm, Jenkins/GitHub Actions, MLflow, Git, AWS (ECS, SageMaker), GCP

Data Infrastructure: PostgreSQL, Neo4j, Kafka, Redis, Iceberg, Elasticsearch, FAISS, MongoDB

PROFESSIONAL EXPERIENCE

MOTIONAL

Pittsburgh, PA

Data Platform Intern (Infrastructure)

May 2026 - Aug 2026

- Enabled characterization of petabyte-scale drivelog data before thawing, by architecting and delivering a metadata-extraction system from PoC to production on a medallion architecture
- Accelerated scene-metadata retrieval and earned a provisional patent, by building an H3 geospatial index and scene embeddings for similarity search
- Delivered natural-language drivelog search and selected the production stack, by building an agentic LangGraph service on FastAPI and benchmarking Ray Data against Spark on Kubernetes

PHENOM

Hyderabad, India

Product Development Engineer (Enterprise Talent Graph)

Dec 2023 - Jul 2025

- Hardened a multi-tenant AI analytics platform for reliability under concurrent load, by leading six engineers to refactor shared services and reduce cross-tenant coupling
- Scaled a production knowledge graph past 1M entities with continuous ingestion, by designing validated aggregation pipelines on Neo4j schemas and Iceberg versioned storage
- Cut AI-driven query response time by 40% across 100+ clients, by tuning PostgreSQL sharding and hardening the Kafka platform (custom exporter, Kafka UI, ZooKeeper-to-KRaft migration)

BYJU'S

Hyderabad, India

Software Engineer (Core Platform Team)

Jul 2022 - Dec 2023

- Cut infrastructure costs by 70% for 1M+ users, by deploying scheduled AWS ECS auto-scaling and Terraform IaC tuned to traffic patterns
- Halved reliance on external identity providers, by leading development of an in-house auth platform (SAML, SSO, RBAC) in Go and Java (Spring Boot)

PHENOM

Hyderabad, India

Machine Learning Intern

May 2021 - Jun 2022

- Boosted recommendation accuracy for production pipelines, by building persona-classification models with spaCy and BERT embeddings across 3 data sources
- Cut recommender preprocessing latency, by migrating CPU pipelines to NVTabular on GPU across 2 production databases

PROJECT EXPERIENCE

FLARE: Efficient Attention (Carnegie Mellon University) (link)

Aug 2025 - Present

- Co-authored a linear-attention operator (3rd of 7 authors, submitted to TMLR) later adopted by an NVIDIA and General Motors crash-dynamics effort for roughly 2x lower memory
- Demonstrated over 200x speedups on Long Range Arena and extended FLARE to image classification and diffusion, by profiling multi-GPU NCCL training across FP8/BF16 with Nsight Systems

Distributed LLM Fine-Tuning with Megatron-LM (Carnegie Mellon University)

Aug 2025 - Oct 2025

- Fine-tuned a 7B-parameter LLM to stable convergence with about 40% less memory on four H100 GPUs, by configuring tensor and pipeline parallelism, gradient checkpointing, and LoRA/QLoRA adapters
- Raised training throughput to 2.1x over naive data parallelism, by tuning NCCL, gradient bucketing, and optimizer-state sharding

CT-Transformer, Vision Transformers (Carnegie Mellon University)

Sep 2025 - Dec 2025

- Reached 96.22% macro AUC-ROC on 15-class VinBigData chest X-ray classification (beating ResNet-50, EfficientNet-B3, and ViT baselines), by training a 210M-parameter Swin Transformer
- Cut inference latency by 3x versus PyTorch eager, by converting to TensorRT (FP16/INT8), accelerating preprocessing with DALI, and serving on Triton with adaptive batching

AWARDS & HONORS

Phenomenal Award for Outstanding Performance, Phenom (2024, Q1 & Q2) • 2nd Place, AppLovin Track, TartanHacks 2026 • Provisional Patent, Motional (2026)